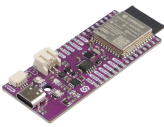


NULA DeepSleep ESP32-S3

SKU 333352



What it is

The NULA DeepSleep ESP32-S3 is a low-power development board with 7 μ A deep sleep current, ideal for battery-powered IoT applications.

Overview

The **NULA DeepSleep ESP32-S3** is a powerful development board engineered for ultra-low-power IoT applications. Built around the **ESP32-S3-WROOM-1-N8R8** module, it combines a dual-core processor with 8 MB of Flash and 8 MB of PSRAM, providing ample resources for complex projects. The board's standout feature is its exceptionally low deep sleep current of just 7 μ A, making it an ideal choice for battery-powered devices where power efficiency is crucial. With integrated **Wi-Fi** and **Bluetooth LE** connectivity, the NULA DeepSleep is the perfect center-stage for your embedded projects! Due to its ample connectivity, it excels in general use applications as well, such as display driving and wireless communicating.

It features an integrated **PCF85063A** real-time clock (RTC) for precise timekeeping, even when the main processor is in deep sleep. Power management is handled by an onboard **TP4056M Li-Ion/Li-Po** battery charger, allowing for battery integration and charging. The board's design ensures stable performance and provides all the necessary components for a long-lasting IoT solution right out of the box. A **USB-C to USB-C** cable is included in the package, so you can start programming immediately without any additional accessories.

Power your next project with the NULA DeepSleep ESP32-S3 if you are building remote environmental sensors, asset trackers, or low-power wireless nodes. This board provides the performance and efficiency you need with its breadboard-friendly layout that makes prototyping quick and easy, while the rich feature set supports a wide range of applications.

Technical details

- ESP32-S3-WROOM-1-N8R8 module (8 MB Flash, 8 MB PSRAM)
- **Ultra-low deep sleep current:** 7 μ A
- Dual-core Xtensa LX7 CPU with Wi-Fi and Bluetooth LE
- USB C to C cable included
- Integrated PCF85063A Real-Time Clock (RTC)
- Onboard TP4056M Li-Ion/Li-Po battery charger
- Breadboard-compatible form factor
- Ideal for battery-powered IoT, data logging, and remote sensing
- qwiic compatible for easy peripheral connection

Specifications

MCU

MCU Part Number	ESP32-S3
MCU Clock	240 MHz
SRAM	512 KB
Flash	4 MB

Connectivity

Wi-Fi	Wi-Fi 4
Bluetooth	Classic + BLE
Wireless Module	ESP32-S3-WROOM-1 (N8R8)

Interface

qwiic Compatible	Yes
Breadboard Compatible	Yes

Power

Battery Connector	JST 2-pin
Onboard Battery Charger	Yes
Supply Voltage	3.3–5 V
Logic Level	3.3 V
Sleep Current	16 μA

Other

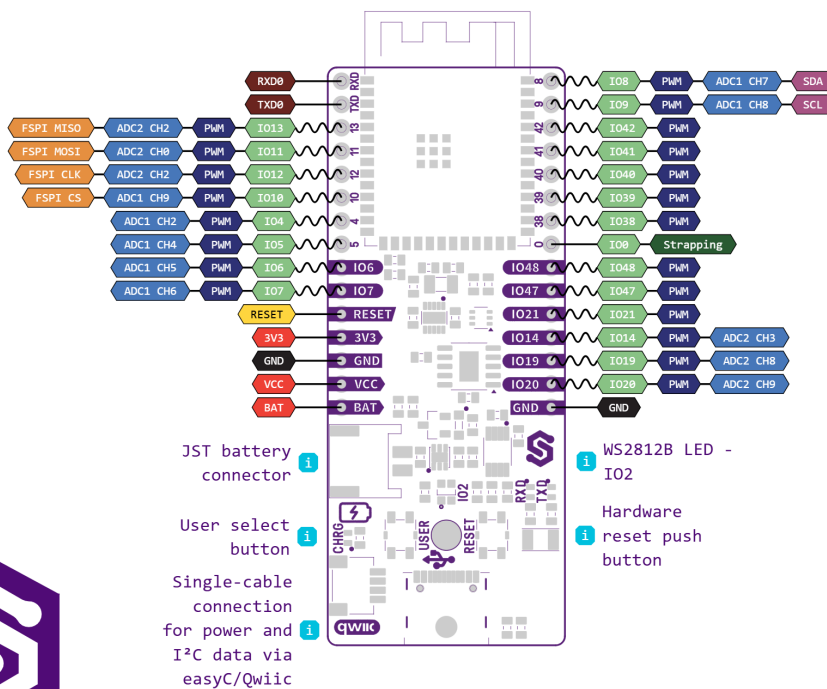
Dev Environments	Arduino IDE, MicroPython, ESP-IDF
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Pinout

SOLDERED NULA ESP32-S3 LOW-POWER - Pinout

Full product documentation:
solde.red/333352

JP1 (NC) - Disconnect WS2812B LED power supply



v1.0

- Pin description
- Power supply pin
- Ground
- Control
- GPIO
- ADC pin
- DAC pin
- SPI comms
- UART comms
- I2C comms
- PWM comms
- Other
- Port name
- General info
- Warning!



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Typical applications

NULA DeepSleep ESP32-S3 is built for low-power wireless firmware development with the ESP32-S3, especially where periodic wakeups and sensor readouts matter. Use it in battery-backed data loggers, remote controls or interactive gadgets that need Wi-Fi and BLE, with the option to sleep deeply between network activity.

What's in the box

- 1 × NULA DeepSleep ESP32-S3 (main product)
- 1 × Header pins (unsoldered) — Header pins included for all breadboard-compatible pins
- 1 × USB-C cable

Variants

- No Headers — SKU 333352
- Male Pin Headers — SKU 333369

Resources

Technical Resources Hardware Details [WEB](#)

Documentation Full Documentation [DOCS](#)

Product data last updated 2026-06-29. Source: solde.red/333352